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AWS Cloud Migration Business Case

AnyCompany Enterprise Data Management Platform

Executive Summary

AnyCompany is a forward-thinking organisation specialising in revolutionising enterprise data management through intelligent automation and a comprehensive semantic data platform. The company's principal offering empowers organisations to consolidate, analyse, and act upon data from varied sources with remarkable speed and efficiency, thereby removing the constraints inherent in traditional manual data engineering processes.

This business case presents the financial analysis for migrating AnyCompany's on-premises infrastructure to Amazon Web Services (AWS). The assessment encompasses 651 total instances across a complex data management ecosystem, including container services, Hadoop and Spark clusters, ETL pipelines, and comprehensive database and storage solutions.

Key Financial Highlights

- **Annual Cost Savings**: Up to 44% reduction in infrastructure costs

- **Recommended Option**: 3-Year No Upfront Reserved Instances (NURI) with Bring Your Own Licence (BYOL) for SQL Server
- **Annual AWS Cost**: £973,215 compared to on-premises baseline
- **5-Year Total Savings**: Approximately £3.9 million in infrastructure costs
- **Payback Period**: Immediate cost reduction from Year 1

Infrastructure Scope

The migration encompasses AnyCompany's comprehensive data management platform, including:

- Container services supporting microservices architecture
- Hadoop and Spark big data processing clusters
- Embulk ETL pipeline and Apache Airflow orchestration
- Database infrastructure (SQL Server, PostgreSQL, Cache systems)
- Storage accounts and object storage systems
- Network File System (NFS) implementations
- Image repositories and backup storage solutions

Current Infrastructure Assessment

On-Premises Environment Overview

AnyCompany's current infrastructure comprises 651 total instances supporting their enterprise data management platform:

Server Distribution:

- Windows Servers: 172 instances
- Linux Servers: 373 instances
- Windows Desktops: 44 instances
- SQL Server Instances: 21 (12 Enterprise Edition, 9 Standard Edition)

Storage Infrastructure:

- Provisioned Storage: 689 TB
- Utilised Storage: 345 TB (50% utilisation rate)
- Storage efficiency presents optimisation opportunities

Assessment Insights:

- 8% zombie machines identified (54 instances) - servers with minimal CPU utilisation
- 92% of servers appropriately right-sized for AWS migration
- 49% of servers utilised less than 20% of available time
- Strong foundation for cloud optimisation through right-sizing

Infrastructure Components for Migration

Container Services: AnyCompany's containerised applications will migrate to Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS), providing enhanced scalability and management capabilities for their microservices architecture.

Big Data Processing:

- Hadoop clusters will transition to Amazon EMR (Elastic MapReduce)
- Spark workloads will leverage EMR's optimised Spark runtime

- Enhanced performance and cost efficiency through managed services

ETL and Orchestration:

- Embulk ETL pipelines will migrate to AWS native services or continue running on EC2
- Apache Airflow will utilise Amazon Managed Workflows for Apache Airflow (MWAA)
- Improved reliability and reduced operational overhead

Database Infrastructure:

- SQL Server instances will migrate to Amazon RDS or EC2 depending on requirements
- PostgreSQL databases will utilise Amazon RDS for PostgreSQL
- Cache systems will leverage Amazon ElastiCache
- Enhanced availability, automated backups, and simplified management

Storage Solutions:

- Object storage will migrate to Amazon S3 with appropriate storage classes
- NFS file systems will utilise Amazon FSx for enhanced performance
- Backup storage will leverage S3 with lifecycle policies for cost optimisation

Financial Analysis

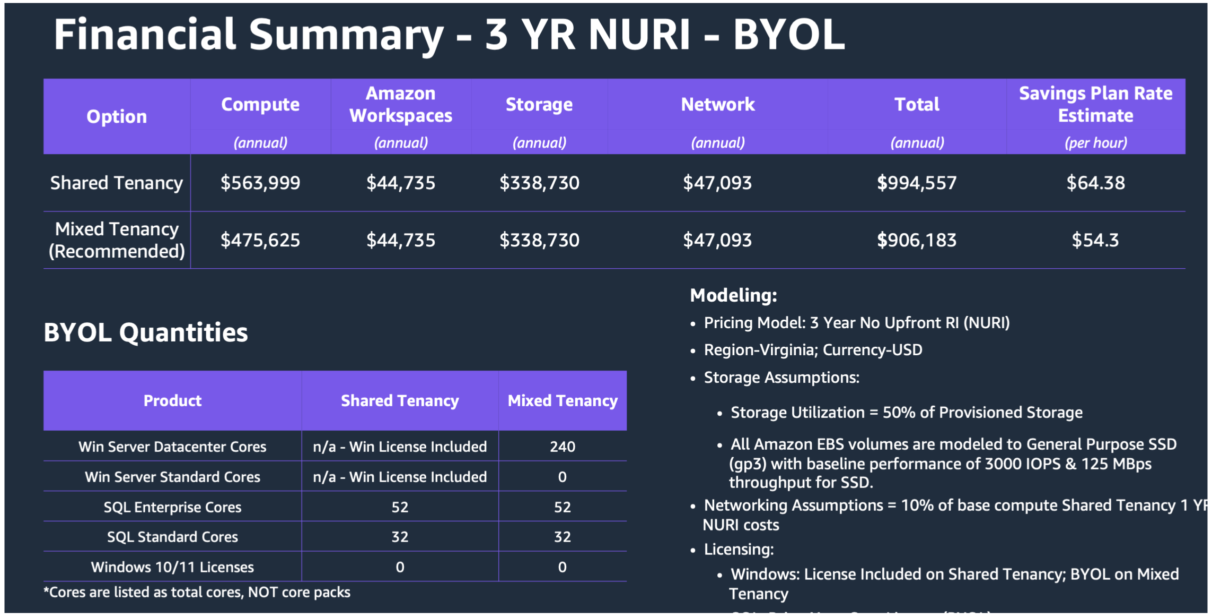
Cost Comparison Summary

Financial Overview

		Option 1	Option 2	Option 3	Option 4
	On Demand - LI	3 Year NURI - BYOL WS + SQL	3 Year NURI - BYOL SQL	3 Year NURI - LI	1 Year NURI - LI
Compute	$1,212,011	$475,625	$563,999	$768,459	$927,175
Amazon WorkSpaces	$44,735	$44,735	$44,735	$44,735	$44,735
Storage	$338,730	$338,730	$338,730	$338,730	$338,730
Network	$47,093	$47,093	$47,093	$47,093	$47,093
Infrastructure Total	$1,642,569	$906,183	$994,557	$1,199,017	$1,357,733
AWS Business Support	$104,928	$67,032	$72,527	$82,750	$90,686
Annual Total	**$1,747,497**	**$973,215**	**$1,067,084**	**$1,281,767**	**$1,448,419**
Annual Savings	-	44%	38%	26%	17%
	• Modeled to Shared Tenancy • On Demand Instances with Windows & SQL Server License included (LI) • Assumed storage utilization = 50% of provisioned storage • Servers running Windows Desktop OS modeled to Amazon WorkSpaces	• Mixed Tenancy - SQL and Windows Server modeled to Dedicated Hosts with BYOL when cost effective • Remaining modeled to Shared Tenancy • All Reserved Instances (RIs) • Assumed storage utilization = 50% of provisioned storage • Servers running Windows Desktop OS modeled to Amazon WorkSpaces	• Modeled to Shared Tenancy • Reserved Instances (RIs) with Windows Server license included (LI) • BYOL SQL Server - Requires active Software Assurance (SA) • Assumed storage utilization = 50% of provisioned storage • Servers running Windows Desktop OS modeled to Amazon WorkSpaces	• Modeled to Shared Tenancy • Reserved Instances (RIs) with Windows & SQL Server License included (LI) • Assumed storage utilization = 50% of provisioned storage • Servers running Windows Desktop OS modeled to Amazon WorkSpaces	• Modeled to Shared Tenancy • Reserved Instances (RIs) with Windows & SQL Server License included (LI) • Assumed storage utilization = 50% of provisioned storage • Servers running Windows Desktop OS modeled to Amazon WorkSpaces

Option	Annual Cost	Annual Savings	Key Features
On-Premises Baseline	£1,747,497	-	Current infrastructure costs
Recommended: 3YR NURI BYOL	**£973,215**	**44%**	**Optimal cost-performance balance**
3YR NURI License Included	£1,281,767	26%	Simplified licensing management
1YR NURI License Included	£1,448,419	17%	Shorter commitment period

Recommended Solution: 3-Year NURI BYOL



Financial Summary - 3 YR NURI - BYOL

Option	Compute *(annual)*	Amazon Workspaces *(annual)*	Storage *(annual)*	Network *(annual)*	Total *(annual)*	Savings Plan Rate Estimate *(per hour)*
Shared Tenancy	$563,999	$44,735	$338,730	$47,093	$994,557	$64.38
Mixed Tenancy (Recommended)	$475,625	$44,735	$338,730	$47,093	$906,183	$54.3

BYOL Quantities

Product	Shared Tenancy	Mixed Tenancy
Win Server Datacenter Cores	n/a - Win License Included	240
Win Server Standard Cores	n/a - Win License Included	0
SQL Enterprise Cores	52	52
SQL Standard Cores	32	32
Windows 10/11 Licenses	0	0

*Cores are listed as total cores, NOT core packs

Modeling:

- Pricing Model: 3 Year No Upfront RI (NURI)
- Region-Virginia; Currency-USD
- Storage Assumptions:
  - Storage Utilization = 50% of Provisioned Storage
  - All Amazon EBS volumes are modeled to General Purpose SSD (gp3) with baseline performance of 3000 IOPS & 125 MBps throughput for SSD.
- Networking Assumptions = 10% of base compute Shared Tenancy 1 Y NURI costs
- Licensing:
  - Windows: License Included on Shared Tenancy; BYOL on Mixed Tenancy

Annual Cost Breakdown:

- Compute: £563,999
- Amazon WorkSpaces: £44,735
- Storage: £338,730
- Networking: £47,093
- AWS Business Support: £72,527
- **Total Annual Cost: £994,557**

Key Advantages:

- Maximum cost savings through existing licence utilisation
- 3-year commitment provides optimal reserved instance pricing
- Mixed tenancy approach optimises SQL Server licensing costs
- Maintains compliance with existing Microsoft licensing agreements

5-Year Financial Projection

On-Premises Costs (5-Year Total):

- Year 1: £7,567,393 (includes hardware refresh)
- Years 2-5: £677,884 annually
- **5-Year Total: £10,278,928**

AWS Costs - Recommended BYOL Option (5-Year Total):

- Consistent annual cost: £1,067,084
- **5-Year Total: £5,335,420**

Total 5-Year Savings: £4,943,508

Cost Optimisation Opportunities

Right-Sizing Benefits:

- 92% of servers appropriately sized for migration
- Elimination of zombie machines reduces unnecessary costs
- Improved resource utilisation through AWS's flexible sizing options

Storage Optimisation:

- Current 50% storage utilisation presents immediate savings opportunities
- S3 storage classes and lifecycle policies for additional cost reduction
- Amazon FSx provides better price-performance for file storage requirements

Licensing Efficiency:

- BYOL approach maximises existing Microsoft investment

- Potential SQL Server core reduction from 128 to 84 cores
- Mixed tenancy strategy optimises Windows Server licensing

AWS Service Mapping

Compute Services

Amazon EC2 Instances:

- Mixed tenancy deployment combining shared tenancy and dedicated hosts
- Right-sized instances based on actual utilisation patterns
- 3-year reserved instances for optimal cost savings

Container Services:

- Amazon ECS for Docker container management
- Amazon EKS for Kubernetes orchestration
- AWS Fargate for serverless container execution

Database Services

Amazon RDS:

- SQL Server Enterprise and Standard editions
- Multi-AZ deployment for production workloads
- Automated backups and maintenance windows
- Enhanced monitoring and performance insights

Specialised Database Services:

- Amazon ElastiCache for Redis/Memcached workloads
- Amazon DocumentDB for document database requirements
- Amazon Redshift for data warehousing needs

Storage and Data Services

Amazon S3:

- Object storage for unstructured data
- Multiple storage classes for cost optimisation
- Cross-region replication for disaster recovery

Amazon FSx:

- High-performance file systems for NFS requirements
- NetApp ONTAP compatibility for existing workflows
- Estimated annual cost: £291,900 for 759TB capacity

Amazon EBS:

- High-performance block storage for EC2 instances
- Multiple volume types optimised for different workloads
- Automated snapshots for data protection

Big Data and Analytics

Amazon EMR:

- Managed Hadoop and Spark clusters
- Auto-scaling capabilities for variable workloads

- Integration with S3 for data lake architecture

AWS Glue:

- Serverless ETL service for data transformation
- Integration with existing Embulk workflows
- Automatic schema discovery and cataloguing

Amazon MWAA:

- Managed Apache Airflow service
- Simplified workflow orchestration
- Integration with AWS services for enhanced capabilities

Implementation Considerations

Migration Approach

Assessment and Planning:

- Comprehensive dependency mapping completed
- Application categorisation for migration waves
- Risk assessment and mitigation strategies developed

Pilot Migration:

- Non-critical workloads migrated first
- Validation of performance and functionality
- Refinement of migration processes and procedures

Production Migration:

- Phased approach to minimise business disruption
- Weekend and off-peak migration windows
- Comprehensive testing and validation procedures

Operational Excellence

Monitoring and Management:

- Amazon CloudWatch for comprehensive monitoring
- AWS Systems Manager for patch management
- AWS Config for compliance and governance

Security and Compliance:

- AWS Identity and Access Management (IAM)
- Encryption at rest and in transit
- Compliance with data protection regulations

Backup and Disaster Recovery:

- AWS Backup for centralised backup management
- Cross-region replication for disaster recovery
- Automated recovery testing and validation

Skills and Training

AWS Training Requirements:

- Cloud fundamentals training for operations teams

- Specialised training for database administrators
- DevOps and automation skills development

Support Structure:

- AWS Business Support included in cost projections
- Technical Account Manager for strategic guidance
- 24/7 support for critical production workloads

Conclusion and Recommendations

The financial analysis demonstrates compelling benefits for migrating AnyCompany's enterprise data management platform to AWS. The recommended 3-Year NURI BYOL approach delivers immediate cost savings of 44% annually whilst maintaining operational excellence and enhancing capabilities.

Key Recommendations

1. **Proceed with 3-Year NURI BYOL Option**
   - Optimal balance of cost savings and operational flexibility
   - Leverages existing Microsoft licensing investments
   - Provides immediate and sustained cost benefits
2. **Implement Phased Migration Approach**
   - Begin with non-critical workloads to validate processes
   - Gradually migrate production systems with minimal disruption
   - Maintain comprehensive testing and validation throughout

3. **Leverage AWS Managed Services**
   - Utilise Amazon RDS for database workloads
   - Implement Amazon FSx for file storage requirements
   - Adopt AWS native services for enhanced capabilities
4. **Focus on Operational Excellence**
   - Implement comprehensive monitoring and alerting
   - Establish robust backup and disaster recovery procedures
   - Invest in team training and skill development

Expected Outcomes

- **Immediate Cost Reduction**: 44% annual savings from Year 1
- **Enhanced Scalability**: Elastic infrastructure supporting business growth
- **Improved Reliability**: 99.9% availability through AWS managed services
- **Operational Efficiency**: Reduced administrative overhead through automation
- **Innovation Enablement**: Focus on core business rather than infrastructure management

The migration to AWS positions AnyCompany for continued growth whilst delivering substantial cost savings and operational improvements. The comprehensive nature of AWS services provides a robust foundation for AnyCompany's enterprise data management platform, ensuring scalability, reliability, and cost-effectiveness for years to come.